

Cold water-flooding in a heterogeneous high-pour-point oil reservoir using computerized tomography scanning: Characteristics of flow channel and trapped oil distribution

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ABSTRACT

There are relatively few studies that consider the influence of reservoir heterogeneity on reservoir wax deposition. This study conducts a computerized tomography scan to study the flow channels, heavy component retention, displacement efficiency, and trapped oil distribution in a high-pour point oil reservoir with a positive sedimentary rhythm under various water injection temperatures. This study greatly improves the understanding of the wax deposition influences on the flow channel. Three displacement models are proposed: water-flooding breakthrough model (49.5 °C), approximate-equilibrium piston-like displacement model (54.5 °C/59.5 °C), and non-equilibrium piston-like displacement model (64.5 °C). When the injection temperature is lower than the wax appearance temperature (59.5 °C), an elliptical wax deposition is formed at the inlet, the most serious deposition is in the middle-permeability layer. A wax deposition barrier is formed away from the inlet, and constrains the downward movement trend of the water injected above. Therefore, the moderate wax precipitation will improve the oil recovery from the top low-permeability layer. However, the heterogeneity in the spaces and strength of the barrier result in the oil recovery from the 100 mD layer at 59.5 °C begin higher than that at 54.5 °C; moreover, at 59.5 °C, the recovery at the inlet is better than that at the outlet. The vortex phenomenon increases the oil recovery from wax deposition area in the bottom layer at the inlet. The heavy component retention is significant, and the oil recovery decreases with decreasing injection temperature. After flooding, the formation mechanisms of the trapped oil include wax deposition, weak water washing, and reservoir sedimentary rhythms. The research results provide initial insights for the evaluation of cold damage.

1. Introduction

Crude oil with wax content greater than 20% and freezing point higher than 40 °C is called high-pour-point (high-PPT) oil (Fu et al., 2013). High-PPT oil is widely distributed worldwide. It is an important source of chemical raw materials, and is widely used in food, medicine, electronics, aerospace, and other fields (Mu et al., 2015). The Shenyang Oilfield, in northeast China is a typical high-PPT oilfield. The formation

structure of the reservoir is positive rhythm, and the formation temperature is higher than the wax appearance temperature (WAT) of the original crude oil by more than 10 °C. The sedimentary rhythm comprises the regular alternation of sediments from coarse to fine or from fine to coarse. Generally, a reservoir whose lithology changes from coarse to fine from bottom to top and whose permeability decreases from bottom to top is called a "positive rhythm reservoir". The serious heterogeneity and special physical properties of crude oil make the flow

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of oil and water more complicated. Production data has shown that most injection wells have high injection pressures in the early production stage; the utilization rate of the injected water gradually deteriorates (owing to the water channel) after a long period of cold-water flooding. These problems result in a new challenge to the previously neglected phenomenon of wax deposition in rock.

High-PPT oil is extremely sensitive to temperature (Hammami and Raines, 1997). When the temperature is lower than the WAT, the long-chain alkanes in the crude oil are separated from the disordered phase and form wax crystals in the solid phase (Fei Yang et al., 2015). The mechanism of wax precipitation is extremely complex and is generally divided into three stages: nucleation, growth and reunion (Vignati et al.; Machado and Lucas, 1999). Wax depositions in the wellbores of production wells and in crude oil transportation pipelines are the most common phenomena. The main cause is the larger temperature difference between the hot oil and cold pipe wall (Leontaritis and Geroulis, 2011). Many studies have focus on restarting the gelled crude oil in the pipeline or wax deposition models at the pipe wall, along with the wax removal force and wax removal efficiency (Becker, 1991; Alboudwarej et al., 2006; Negrão et al., 2011; Kumar et al., 2015; Kumar et al., 2016; Sun et al., 2016a,b; Chala et al., 2018; W. Li et al., 2019).

However, relatively little attention has been devoted to the wax deposition in pores. Although the wax deposition in sandstone was discovered very early (Brewster, 1927; Reistle, 1927) and was found to restrict fluid flows (Brewster, 1927; Nanniger and Nanniger, 1990), insufficient attention has been paid there to by researchers and producer. It is often difficult to recognize the presence of wax deposits in pores during the development process. Reistle (1927) proposed three wax deposition issues affecting crude oil production, including wax deposition in the wellbore, wax deposition in the sandstone and wax deposition in oil pipelines, but no in-depth analysis was conducted for each issue. Maloney and Osthus (2005) used environmental scanning electron microscopy to obtain present evidence of paraffin in pore system. They found that the wax distribution was heterogeneous, and that its relative abundance was related to the initial oil saturation before cooling. In the development of the Dam Joao high-PPT oilfield in Brazil, the researchers proposed that wax deposition is the most important factor leading to declines in production, rather than the previously proposed reservoir heterogeneity, after improving the oil recovery by injecting hot steam (Romeu et al., 1990; Mezzomo and Rabinovitz, 1999, 2001).

Some recent studies have discussed the wax deposition in pores through experiments and mathematical models. Sutton and Roberts (1974) injected cold water into 4.2-mD sandstone and 412-mD sandstone to study the decrease in permeability caused by wax deposition. The results showed that the permeability of the 4.2-mD core was decreased by 90% after a 0.5 pore volume (PV) water injection, and that a 3.75 PV injection was required to reach the same value in the 412-mD core. Based on the above experimental results, Ring et al. (1994) established a model for describing paraffin precipitation, convection and deposition considering the reservoir thermal effect, and used the experimental data from (Sutton and Roberts, 1974) to fit the model. S. Wang et al. (1999) established a method for considering the effects of static and dynamic pore surface depositions and pore throat plugging on paraffin retention. They believed that pore throat plugging deposition can only occur under specific conditions and verified the model based on the experimental data in (Sutton and Roberts, 1974). Some studies have derived relationships between the amount of paraffin deposition and certain-formation parameters by establishing a one-dimensional capillary model and have proposed that there is a better linear or exponential relationships between the amount of wax deposition and the decrease in core permeability (Xu et al., 1991; Fan Yang and Li, 2006; K. Li et al., 2014; Jiang et al., 2015). Others have reached similar conclusions (K. Li et al., 2014; Nie, 2015). However, some investigations have proposed that the effect of wax deposition in the pores is related to the temperature difference between the WAT of the crude oil and the formation

temperature. If the temperature difference is greater than 10 °C, the cold damage caused by the water injection can be ignored (Mu et al., 2015); however, some field engineers believe that this temperature difference should be increased to 20 °C. The deposition of asphaltenes in pores also has similar effects on wax deposition (S. Wang et al., 1999). However, many studies have shown that asphaltene deposition mainly occurs after the reservoir pressure decreases or after CO₂ injection (Ali and Islam, 1998; Hamadou et al., 2008; Alizadeh et al., 2009; Shen and Sheng, 2018; Davudov and Moghanloo, 2019; Ghadimi et al., 2020): it is less affected by temperature, and thus is the opposite of the main controlling factor of wax deposition. Therefore, there have been relatively few studies on the wax deposition in pores, and most of them have been conducted at the same temperature (L. Wang et al., 2019), i.e., the injection temperature is the same as the core temperature. This is quite different from the actual situation in an oilfield.

The effects of heterogeneity in reservoirs have always received significant attention. For reservoirs with a positive rhythm, the difference in the vertical water injection is enlarged owing to the low permeability in the upper part of the layer, and the injected water flows along the high-permeability layer first under the influence of gravity. Thus, the water-flooding degree in the upper part of the reservoir becomes increasingly worse (Du et al., 2003; X. Liu and Liu, 2006). However, in high-PPT oil reservoirs, cold-water injection inevitably leads to wax deposition in the pores, which will affect the formation of flow channels and distribution of trapped oil. In addition, most high-PPT oils show the characteristics of a light oil at the formation temperature. The characteristics of the cold water injection are also different from those of heavy oil steam flooding, steam assisted gravity drainage, Vapor Extraction and foam flooding (Al-Hadhrani et al., 2014; Lang et al., 2020; P. Liu et al., 2020; Wan et al., 2020). Therefore, further studies are still needed on the flow processes and laws of injected water in a heterogeneous high-PPT oil reservoir.

There are three commonly used methods for studying the flow channel studying: two-dimensional etched glass micromodels, nuclear magnetic resonance (NMR), and core computed tomography (CT) scans (Y. Zhao et al., 2017). Although the two-dimensional etched glass micromodel can be designed according to the pore structure of the actual core, its pressure threshold ranges from 0 to 1 MPa, limiting the development of live oil experiments or constant-speed displacement experiments. The special physical properties, such as the WAT, viscosity, and PPT of high-PPT live oil are quite different from those of dead oil (M. B. Wang et al., 2019). NMR can quantitatively distinguish the core pore structure and fluid distribution for large, medium, and small pores. NMR has developed rapidly in regards to low permeability and tight oil in recent years (Dai et al., 2019; M. Chen et al., 2020); however, owing to the limitations of the heating method and its long scanning time, it is not suitable for high-PPT oil experiments. CT scanning can obtain the fluid distribution of the core in situ; it can even capture the dynamic process and can display the displacement process more accurately. After comparing two CT images before and after oil displacement, T. Wang et al. (2013) found that the asphaltene deposition in sand resulted in a decrease in the permeability of tight sandstone reservoirs after CO₂ injection. Tang et al. (2019) studied a foam flow for enhanced oil recovery with two representative model oils based on dual energy CT. In addition, online CT technology has also been used to study the mechanisms of water imbibition and displacement in conglomerates (Deng et al., 2014; H. Zhao et al., 2019). Therefore, to the best of our knowledge, there has been no investigation on the formation and development of an injected water flow channel under the influence of wax deposition in heterogeneous high-PPT live oil reservoirs.

This study is the first to analyze the influences of wax deposition on the flow channel in a cold water-flooding process, using online CT technology. We used a positive rhythm artificial core with 3% clay content and live oil in the experiment, to replicate the actual formation conditions as much as possible. On the premise of maintaining the core temperature at 59.5 °C, we investigated the formation and development

of flow channels during the water-flooding progress at different temperatures through instantaneous 3D reconstruction and oil saturation distribution profiles. Simultaneously, the pressure drop from inlet to outlet with time was monitored to further analyze the displacement model for the injected water and plugging of the wax crystals. In addition, we also measured the crude oil components in the produced fluid at 0.6 PV and 1.2 PV to analyze the retention of heavy components in the core. Finally, we discussed the displacement efficiency and trapped oil distribution under various water injection temperatures, and the significance of the experimental results in regards to engineering applications.

2. Experimental section

2.1. Experimental fluids

Live oil was used for the experiment to better resemble the real conditions, as the physical property differences with dead oil were larger. The live oil was made up of dead oil taken from the wellhead of the S84A12 high-PPT oilfield, and the dissolved gas was consistent with the dissolved gas composition. The gas-oil ratio was $37 \text{ m}^3/\text{m}^3$, i.e., consistent with the actual formation. Based on a laser method, the WAT of the live oil under the original formation pressure (19.5 MPa) was 59.5°C and the PPT was 41°C , which were 1.6°C and 7.0°C lower, respectively, than those of dead oil at atmospheric pressure. The wax content of the crude oil is 55.18%. The viscosity of crude oil at 19.5 MPa and 70°C was 9.8 mPa s, as determined by the capillary viscometer method. The injected water type was Na_2CO_3 , and the salinity was 3384 mg/L. The detailed parameters are listed in Table 1.

2.2. Experimental cores

The rhythm type of the S84A12 block was positive, and the permeability was distributed from 1 to 2700 mD, with an average permeability of 396 mD. To avoid the changes in the core sample properties and to facilitate comparison of the experimental results, an artificial core was used in the experiment. The core adopted three layers of heterogeneous core with positive rhythms. The designed permeability value from the top to bottom layers was 100 mD, 700 mD and 1200 mD, respectively; the cross-section length was $4.5 \times 4.5 \text{ cm}$; and the cross-sectional areas of the three layers were the same (Fig. 1). According to the previous core preparation and the current artificial core technology level, the permeability of each layer could achieve the design requirements, and the actual gas permeability of the core was 697 mD. During the manufacturing process of the artificial core, 3% clay was added to simulate the water-sensitive characteristics. The results of a clay analysis

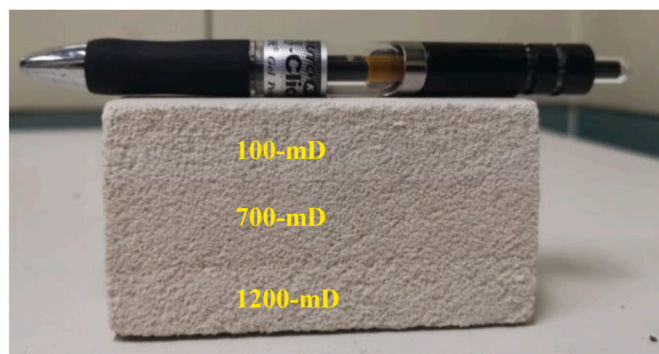


Fig. 1. Three layers of artificial core with length of 10 cm.

and electron microscope analysis showed that the diagenetic minerals were mainly clay minerals and carbonate minerals, and the clay minerals are mainly montmorillonite and illite. The reservoir had weak to medium water sensitivity. When low-salinity water was injected, the expansion of the clay led to a decrease in the reservoir permeability and the accumulation of broken clay particles in the pore throat to form blockages; in addition, the expansion of the clay occupied the original space of crude oil, potentially leading to increases in the wax precipitation rate and thickness of the wax deposition layer.

2.3. Experimental apparatus

2.3.1. Online computerized tomography (CT) system

The CT scanning system (Fig. 2.) was a Lightspeed 8-layer spiral CT scanner, with a tube voltage of 100 kV, tube current of 100 mA, scanning layer thickness of 2.5 mm, scanning layer interval of 10 mm, four detector rows, scanning time frequency of 0.5s/3600s, and an image reconstruction time of 0.167 s (512×512 matrix). The CT scan data were reconstructed and calculated using image processing software. The coreholder shell was made of polyether-ether-ketone material. The X-ray absorption capability of this material of this material is quiet weak; thus, X-ray can penetrate the core sample smoothly, thereby reducing the measurement errors caused by the ray hardening effect.

2.3.2. Core displacement experimental system

The experimental apparatus of the core displacement system comprised three systems: a displacement system, a heating system, and measurement system, as shown in Fig. 3. The displacement system consisted of two high-precision injection pumps (Quizix SP-5000) for achieving a stable flow rate or pressure balance, one ISCO pump (Tel-edyne A260 ISCO, Thousand Oaks, CA, USA) acting as the confining pressure control system, a live oil tank, kerosene tank, formation water tank, and a core holder made of PEEK material, as well as several pressure gauges, pipelines, and valves. The pressure monitor recorded the pressure at the inlet and outlet ends of the core holder to calculate the pressure drop. The heating system consisted of two independent heating units and a constant-temperature electric heating wire at the end of the outlet. The constant-temperature electric heating wire heated the back-pressure valve and back-section pipeline at a constant temperature of 70°C to ensure the mobility of the high-PPT oil and to avoid the solidification of crude oil in the pipeline caused by the rise of the WAT and PPT after degassing. The measurement system mainly comprised an oil-water separator, gas flowmeter, and data collector. In addition, a high-pressure live oil bottle (Proserve type 6, Houston, TX, USA) was used in the transportation and storage process of the live oil to maintain the pressure stability of the bottle through the elasticity of the high-pressure nitrogen, thereby avoiding the degassing of the crude oil in the bottle caused by the change in the temperature.

Table 1

Physical properties of the dead oil samples.

Crude oil	Parameter	Test method	Value
Dead oil	Density (kg/m^3)	GB/T1884-2000 (20°C , 0.1 MPa)	865.3
	Viscosity (mPa·s)	GB/T 22,235-2008 (70°C , 0.1 MPa)	14.7
	Gas-oil ratio (m^3/m^3)	SY/T 5542-2009	37
	Saturation pressure (MPa)	SY/T 5542-2009	8.7
	Pour point (PPT) ($^\circ\text{C}$)	SY/T 0541-2009	49
	Wax appearance temperature (WAT) ($^\circ\text{C}$)	SY/T 0545-2012 (DSC)	61.43
	Wax content (wt %)	SY/T 0537-2008	55.18
Live oil	WAT ($^\circ\text{C}$)	Laser method (19.5 MPa)	59.5
	PPT ($^\circ\text{C}$)	Laser method (19.5 MPa)	41
	Viscosity (mPa·s)	Capillary viscometer (70°C , 19.5 MPa)	9.8
	Viscosity (mPa·s)	Capillary viscometer (55°C , 19.5 MPa)	40.1



Fig. 2. Schematic diagram of online computerized tomography (CT) system.

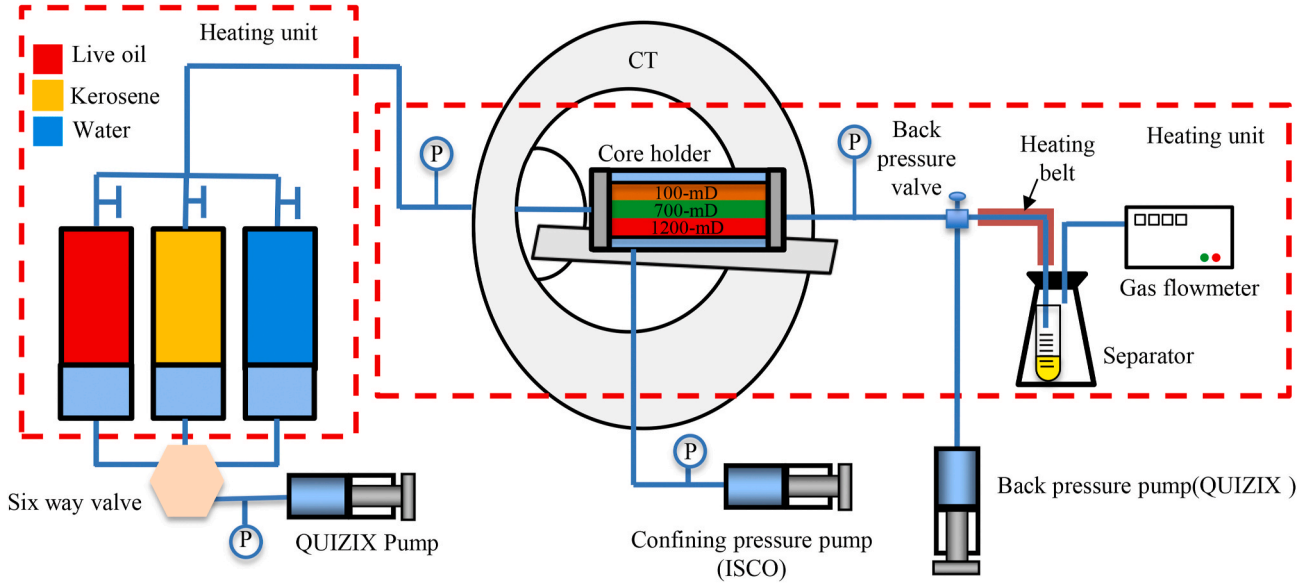


Fig. 3. Schematic diagram of core displacement experimental apparatus.

2.4. CT measurements

By measuring the amount of X-ray emissions, the attenuation coefficient of each unit volume of the slice was calculated after digitization, and the attenuation coefficient matrix was reconstructed on a high-speed computer. Then the image was formed (Lü et al., 2012; Deng et al., 2014). Each pixel in the image also contained a CT value (after the attenuation coefficient conversion). As different materials have different CT values, the different CT values on the core scale could be used to identify the saturations of different fluids. According to $S_w + S_o = 1$, the water saturation S_w and oil saturation S_o could be calculated by combining dry model, wet, and intermediate model (i.e., the scanning data of dry rock fault plane, 100% saturated fluid, and rock fault plane under oil-water coexistence).

$$S_o = \frac{CT_{waterwet} - CT_t}{CT_{waterwet} - CT_{dry}} \frac{CT_{water} - CT_{air}}{CT_{water} - CT_{oil}} \quad (1)$$

$$S_w = 1 - \frac{CT_{waterwet} - CT_t}{CT_{waterwet} - CT_{dry}} \frac{CT_{water} - CT_{air}}{CT_{water} - CT_{oil}} \quad (2)$$

In the above, CT_{dry} is the CT value of the dry rock; $CT_{waterwet}$ is the CT value of the rock after 100% saturated water; CT_t is the CT value of the rock at the time of the intermediate model; CT_{water} is the CT value of the water; CT_{oil} is the CT value of the crude oil; CT_{air} is the CT value of the air.

To better distinguish between water and oil, it is common practice to add NaI to the water or iodo-hexane to the oil as a CT enhancer, so as to expand the CT value comparison between the oil and water phases. Owing to the uncertain influence of adding NaI to the formation water on the physical properties of rock and subsequent alkaline/surfactant/polymer (ASP) flooding, 5 wt% iodo-hexane was added to the live oil, according to experience. Temperature also affects the CT values (Imai et al., 2019). The CT values of the liquids under various conditions are shown in Fig. 4. Adding only 5 wt% iodo-hexane to the oil phase can increase the contrast between the oil and water phases, and the CT value

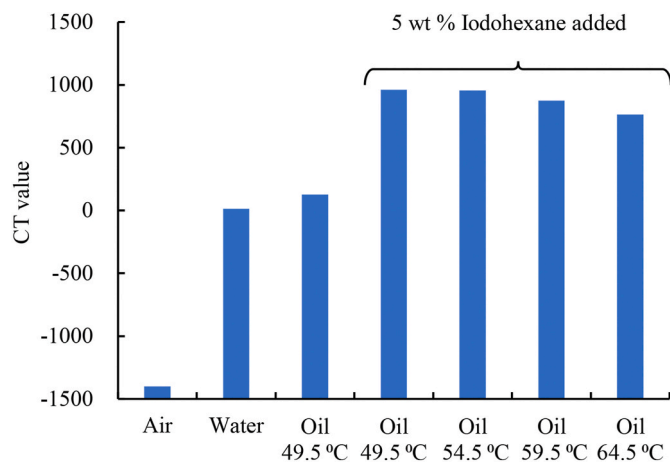


Fig. 4. CT values of air, water and oil under various conditions.

of crude oil increases with decreasing temperature. We used the test results to modify CT_{oil} under various temperatures in Equation (1).

2.5. Experimental procedure

The detailed operation process of the online CT displacement experimental procedure is as follows. **Step 1:** A CT scan was conducted for the live oil, formation water, and air to obtain the CT values. **Step 2:** The live oil in the high-pressure live oil bottle was transferred to the oil tank before the displacement experiments to ensure that the live oil was not degassed during the transportation. **Step 3:** The oil tank, kerosene tank, water formation tank, core holder, pipelines and valves were heated to 70 °C and kept there for 24 h. **Step 4:** The core samples were evacuated for 1 h, and a CT scan was performed to obtain the CT_{dry} model; then, the samples were saturated with formation water for 2 h under 0.1 MPa, and a CT scan of the 100% water-saturated samples was conducted to obtain the $CT_{waterwet}$ model. **Step 5:** The kerosene valve was opened to flush the core from Step 4 at a rate of 0.5 mL/min until no water was produced; then, the flow rate was gradually increased to 10 mL/min until the 10 PV reached to get the connate water saturation. **Step 6:** The back pressure was set to 19.5 MPa, and the system pressure was gradually increased to 19.5 MPa with kerosene. The confining pressure was correspondingly gradually increased to 27 MPa, and then was kept constant. **Step 7:** The live oil valve was opened to inject 3 PV of

live oil into the core. **Step 8:** The temperatures of core holder and pipeline were reduced to 59.5 °C (WAT) at 19.5 MPa, and the temperature of the formation water tank was reduced to the design temperature (49.5 °C). At the end of cooling, the constant temperature was maintained for 12 h to ensure the pressure and temperature balance of the displacement system. Then, a CT scan was performed to obtain the CT_t ($t = 0$). **Step 9:** The core was flushed at a flow rate of 0.8 mL/min with 1.2 PV of formation water until there was no more oil production, and the a CT scan was taken every 4 min to obtain the CT value CT_t . **Step 10:** Steps 3–9 were repeated for the other core samples under various conditions, and the temperature of the core holder was maintained at 59.5 °C; however, the injection temperature was adjusted to 54.5 °C, 59.5 °C (WAT), and 64.5 °C, respectively.

Finally, after each group of experiments, the oil samples were collected and sent to the State Key Laboratory of Heavy Oil Processing to detect the component changes in the crude oil. Owing to the extremely high water-cut content in the later-stage produced liquid as shown in Fig. 5, the produced liquid was generally heated at 70 °C in a sealed tube so to prevent the components from evaporating. Then, cooling was conducted at 20 °C before collecting the surface oil film. In addition, all the collected oil films were sealed and heated again, and sent for testing after dehydration.

3. Results and discussion

3.1. Rock physical distribution and heterogeneity

3.1.1. Porosity distribution

Before the displacement test, it is necessary to verify the reliability of the CT parameters and results. The average porosity of core #1 as obtained by the CT scanning was 0.289, which is basically consistent with the porosity (0.297) measured using water. Fig. 6 shows the distribution of the porosity in the CT scan slices in different directions. We adjusted the porosity color scale to 0.3–0.4 to show the results more clearly.

Fig. 6a shows that the number and density of the colors from the top to bottom increase in the layers, indicating that the core #1 has a positive rhythmic distribution. Fig. 6b shows the axial porosity distribution of each layer. There are evident layer differences in the color display. In addition, owing to the high permeability of the core, the porosity curve at the outlet suddenly drops in the range of 97 mm–100 mm. However, from comparing the axial core porosity distribution, it is observed that the core average porosity in each layer changes little along the axial direction, except at the outlet end.

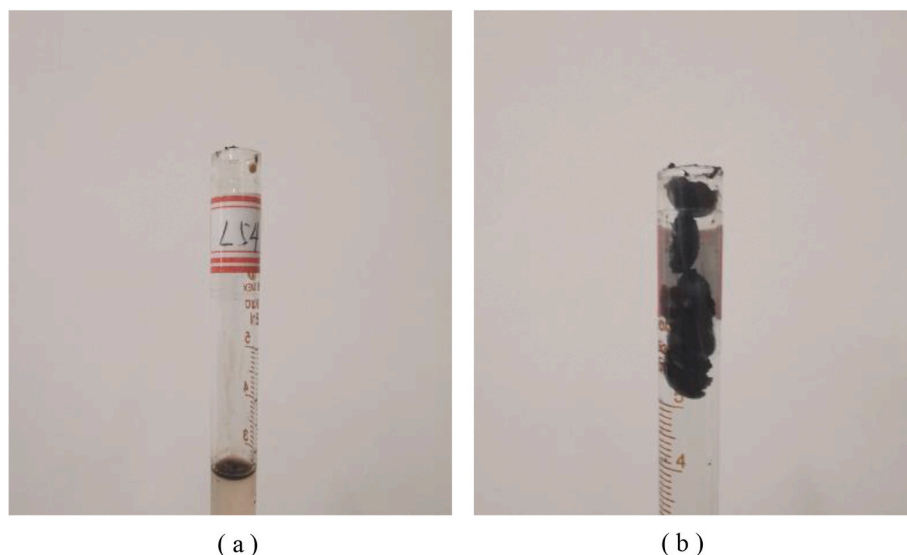


Fig. 5. Collection of test oil samples: (a) Oil film formed after heating and cooling; (b) Multiple oil film collection.

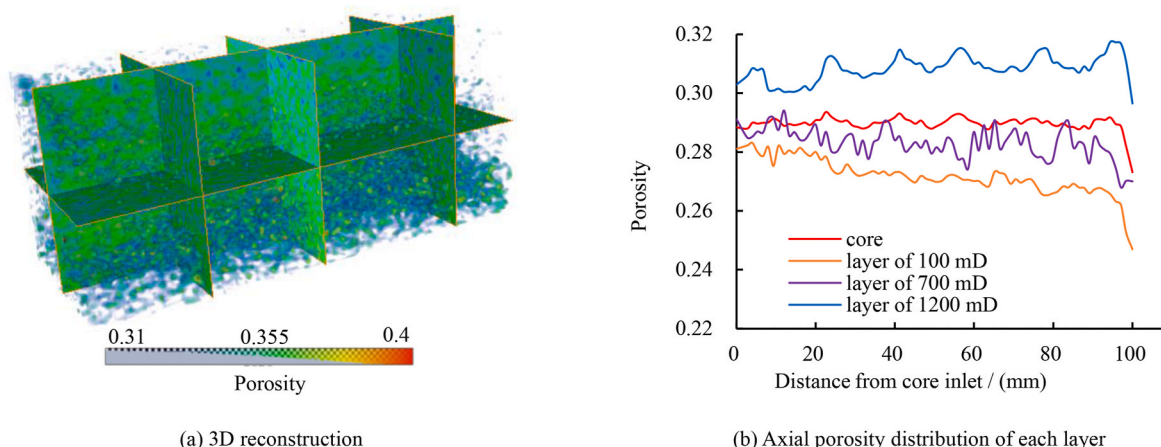


Fig. 6. Porosity distribution of Core #1.

3.1.2. Oil saturation distribution under connate water saturation

The next step is to evaluate the maximum oil saturation (S_{omax}) under connate water saturation in the different layer. First, the CT_t ($t = 0$) of core #1 was substituted into Equation (1), to calculate the S_{omax} value at each pixel. Simultaneously, the threshold value of oil saturation was set from 0 to 1. Then, the arithmetic average method was used to obtain the S_{omax} at each slice. The S_{omax} of core #1 from the same method was 83.1%. The results are basically consistent with the oil saturation value of 82.2% obtained by the oil-water measurement method, demonstrating the reliability of the CT scan results.

Fig. 7a shows that the crude oil (red) is distributed in a large area in the continuous phase in most areas. The connate water is distributed in it in blue-green point or strip shapes, indicating that most areas of core #1 are fully saturated. However, in the upper part of the outlet, in the dotted black line frame, the formation water presents a continuous phase, and the width and water saturation increase with decreasing distance from the outlet. This is the most common phenomenon in the crude oil saturation process of a heterogeneous core and is consistent with the experimental expectations. Fully saturating the core with crude oil to achieve the original state has always been our pursuit. Although the method of increasing the flow rate has been adopted to solve the above problems, the effect is still not ideal. Other methods, such as reverse saturation or layers with different permeabilities glued together after saturation, can reduce the water saturation at the outlet to a certain extent. However, they are limited by the connection order of the experimental equipment and the requirements to ensure that the crude oil does not degas; thus, these methods were not used in the experiments.

Fig. 7b shows the actual situations of insufficient saturation in each

layer. Among them, the 700-mD layer and 1200-mD layer show a smaller decline, and the change ranges are generally within 10% (as compared with the oil saturation in the inlet). As for the 100-mD layer, the change of the oil saturation within 60 mm away from the inlet is basically the same as in the other two layers. However, the rate of decline gradually increases from 60 mm, and the oil saturation drops sharply from 75% to 43% within the distance 20 mm from the outlet. The reasons for this phenomenon are as follows.

First, at a low flow rate, most of the oil initially enters the large pores in the 1200-mD layer, i.e., those with a smaller capillary resistance for displacing water. Then, increasing the injection rate leads to two effects. First, the high injection pressure difference causes the crude oil in the 1200-mD layer to overcome the capillary force and enter the secondary large pores, thereby displacing the formation water. Second, the high injection speed expands the oil sweep area, and crude oil successively enters the middle- and low-permeability layers with high capillary resistance. However, owing to the Jamin effect and other additional resistances, the formation water in the medium- and low-permeability layers cannot be completely driven out. Finally, the oil saturation of the 100-mD layer of core #1 reaches 75.42%, i.e., lower than 83.23% for the 700mD-layer and the 87.19% for the 1200mD-layer. Table 2 lists the porosity and oil saturation values of the other three cores and layers. According to the requirements of the experimental design, the water injection temperatures of cores #1 to core #4 are 49.5 °C, 54.5 °C, 59.5 °C (WAT at 19.5 MPa), and 64.5 °C, respectively.

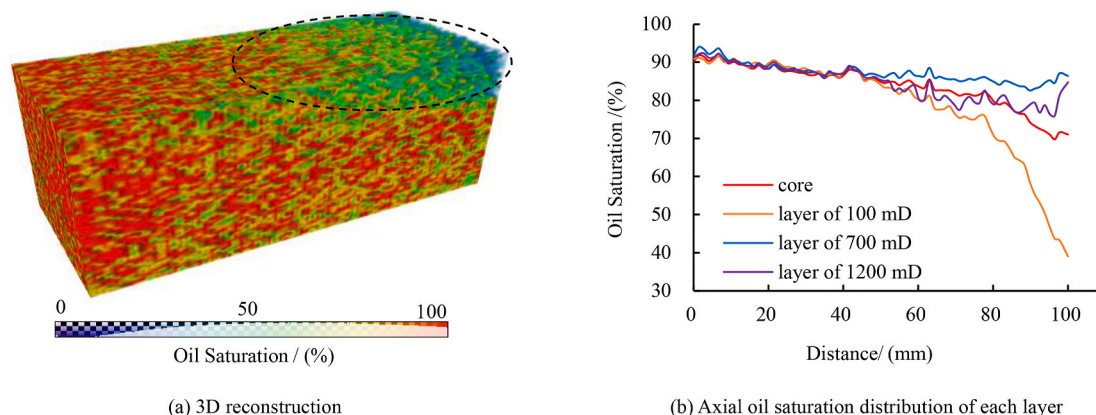


Fig. 7. Oil saturation distribution of core #1.

Table 2

Porosity and oil saturation in different cores.

Core Number	Porosity tested by water	Computerized tomography (CT) Porosity				Experiment oil saturation/(%)	CT oil saturation under connate water saturation/(%)				Design water injection temperature/(°C)
		Core	100 mD	700 mD	1200 mD		Core	100 mD	700 mD	1200 mD	
Core #1	0.297	0.294	0.282	0.292	0.308	83.57	81.95	75.42	83.23	87.19	49.5
Core #2	0.302	0.291	0.284	0.289	0.301	81.52	80.81	72.79	84.01	85.63	54.5
Core #3	0.308	0.298	0.288	0.294	0.313	82.73	82.32	76.71	84.75	85.51	59.5
Core #4	0.308	0.301	0.286	0.303	0.311	81.34	79.75	72.83	81.82	84.62	64.5

3.2. Formation and change of flow channel

3.2.1. Analysis of flow channel through 3D reconstruction

Both the heterogeneity of the reservoir and the wax deposition in the pores will affect the fluid flow. To show the propagation and flow in the core of injected water more clearly, the color legend threshold for crude oil saturation was adjusted from 96% to 100%. To better explain the flow direction, the flow channel was marked in the oil saturation distribution map. For example, the main flow channel 1 of the bottom high-permeability layer was marked as C.1, the axial component of main flow channel 2 was marked as C.2-1. The arrow represents the main flow direction, the solid line represents the main flow channel, the dotted line represents the component of the main flow channel, and the length of the dotted line represents the size of the distributary flow channel.

As shown in Fig. 8, when the water injection temperature is 49.5 °C, the flow channel distribution is quite different from that of a conventional heterogeneous reservoir. The elliptical wax deposition area at the inlet is evident. Most of the cold water enters the medium- and high-permeability layers, causing the high PPT oil temperature to drop rapidly, and precipitating wax. Owing to the larger pore size, the capture capacity of the 1200-mD layer in regards to wax crystals is smaller than that of the 700-mD layer; the wax deposition in the top 100 mD layer is

also lower, as less cold water enters. Therefore, an elliptical wax deposition area protruding toward the outlet with the 700-mD layer as the center is formed at the inlet. This area is formed in a short time, and constrains the expansion of the flow channel. At a constant injection rate, the injected water begins to move slowly along the bottom of the 1200-mD layer with minimal seepage resistance to break through, and gradually forms the main channel #1 (C. 1). The water breakthrough is at a PV injection of 0.1. Furthermore, owing to heat convection between the cold water, oil, and matrix, the injected water temperature rises after passing through the narrow channel at the inlet, reducing the influence of wax blockages. The injected water sweep area in channel #1 expands after reaching far away from the inlet, presenting characteristics of a near-piston-like displacement.

Furthermore, at a 0.2 PV water injection, flow channel #2 (C. 2) in the upper core gradually becomes apparent. Under the combined action of gravity and the displacement pressure, the injected water moves diagonally downward along the upper wax deposition boundary, and forms two directions. One is the flow channel # 2-1 (C. 2-1) along the core axial direction; it is dominated by the displacement pressure, and is the main flow channel in the upper core. Its formation mechanism is similar to that of channel #1, but because the seepage resistance is greater than that of the 1200-mD layer, the formation time is later;

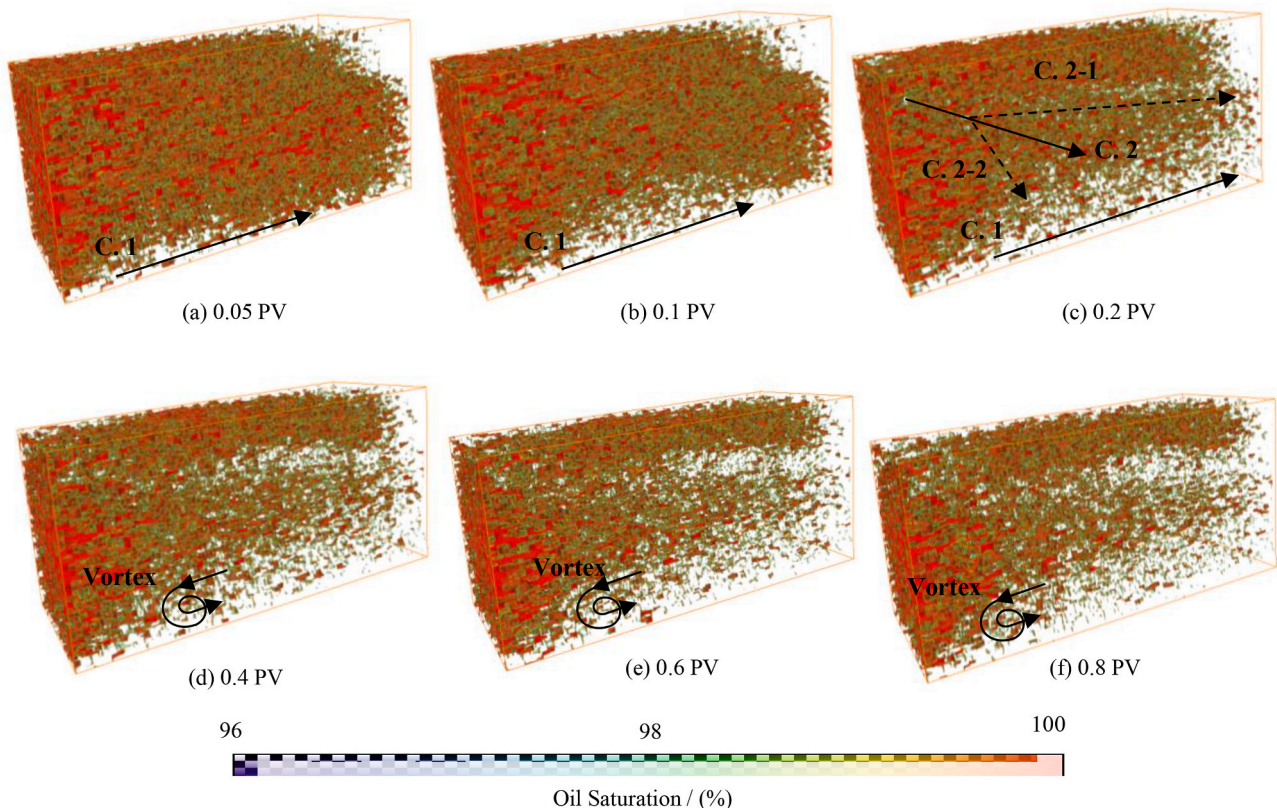


Fig. 8. Oil saturation distribution of various pore volume (PV) water injections at injection temperature of 49.5 °C.

moreover, it is mainly located in the range between the 700-mD upper layer and 100-mD bottom layer. The second is channel # 2-2 (C. 2-2), which is dominated by gravity. The injected water moves downward and converges on channel #1 to further expand the reach of the bottom-injected water. However, compared with the displacement pressure, the influence of gravity is relatively weak, and the flow channels are scattered in the area where the wax deposition blockage is weak. In addition, there are wax deposition barriers between flow channel #1 and channel # 2-1 under a lower injection water temperature, increasing the resistance of the injection water in channel # 2-1 to the downward movement. At this time, the flow channel for the injected water has been formed, and there is no evident change with increasing PV injection.

At 20 mm from the inlet, after a 0.4 PV water injection, the boundary of the wax deposition area in the 1200-mD layer tends to bulge toward the inlet with increasing PV water injection (Fig. 8 d, e, f). Therefore, the oil recovery at intervals of 5 mm within the 1200-mD layer at 20 mm from the inlet is shown in Fig. 9 a. The same results are also shown in Fig. 9 b; however, the above laws are not significant when the water injection temperature is at or higher than the WAT (Fig. 9 c d). The crude oil recovery rate declines from 15 to 20 mm; the oil recovery rate is the largest at the time of the 0.6–0.8 PV in the range of 10–15 mm and then decreases. At 5 mm from the inlet, the maximum crude oil recovery rate is at the time of 0.8–1 PV. The results indicate that the vortex phenomenon in the high-permeability layer leads to an improvement in the oil recovery. As the high-permeability layer is the main flow channel, the flow area of the injected water suddenly expands after passing through the narrow channel formed by wax deposition at a high speed; then, the flow rate slows down, and a part of it forms a vortex opposite to the main flow direction. The wax deposition on the boundary and trapped oil are eroded and broken from the pores and throats under the action of the vortex stress, and are finally carried out by the injected water.

At a 54.5 °C water injection temperature, the initial characteristic of

the flow channel is similar to that at 49.5 °C, that is, channel #1 (C.1) is also formed at the inlet. However, owing to the 5 °C increase in the injection temperature, the decrease in wax deposition results in a significant difference between them. The volume of the wax deposition area is reduced, and the injected water rapidly passes through the short narrow channel at the bottom of the 1200-mD layer. With a 0.05 PV water injection, the upper flow channel slowly forms in the dotted circle above the core. When a 0.1 PV injection is reached, flow channel #2 (C.2) is clearly visible (Fig. 10). Then, the displacement efficiency of the 700-mD and 1200-mD layers is greatly improved. In Fig. 10c, a clear water-flood front can be seen. The displacement process is mainly a near-piston-like displacement. Three reasons lead to an increase in oil recovery, as follows.

First, the wax deposition barrier between flow channel # 2-1 (C. 2-1) and flow channel #1(C. 1) prevents the fluid from flowing downward. Second, the temperature difference between the injected water and crude oil is small, and the injected water temperature increases away from the inlet. The wax precipitation decreases in the crude oil result in a decrease in the oil-water mobility ratio; hence, the injection water displacement front is more uniform. Third, the flow resistance is reduced in flow channel # 2-1 after the water breakthrough, and the injected water in channel #1 is partially distributed to the upper core, expanding the sweep area of the 100-mD layer. When the water injection reaches 0.8 PV, the oil recovery of the upper layer is evident better than that at other water injection temperatures.

Fig. 11 presents the formation and changes of the flow channel when the injection temperature is 59.5 °C. The injected water is still the first to flow along channel #1 (C. 1) in the 1200-mD layer. However, the flow channel changes again. Notably, at the WAT, a small amount of wax crystals have been precipitated (Hammami and Raines, 1997; M. B. Wang et al., 2019). Hence, the smaller wax deposition causes the resistance of wax deposition in the middle-permeability layer to be greatly reduced relative to that at the lower temperature, and the flow channel #2 in the 1200-mD layer quickly forms. The water-flood front is

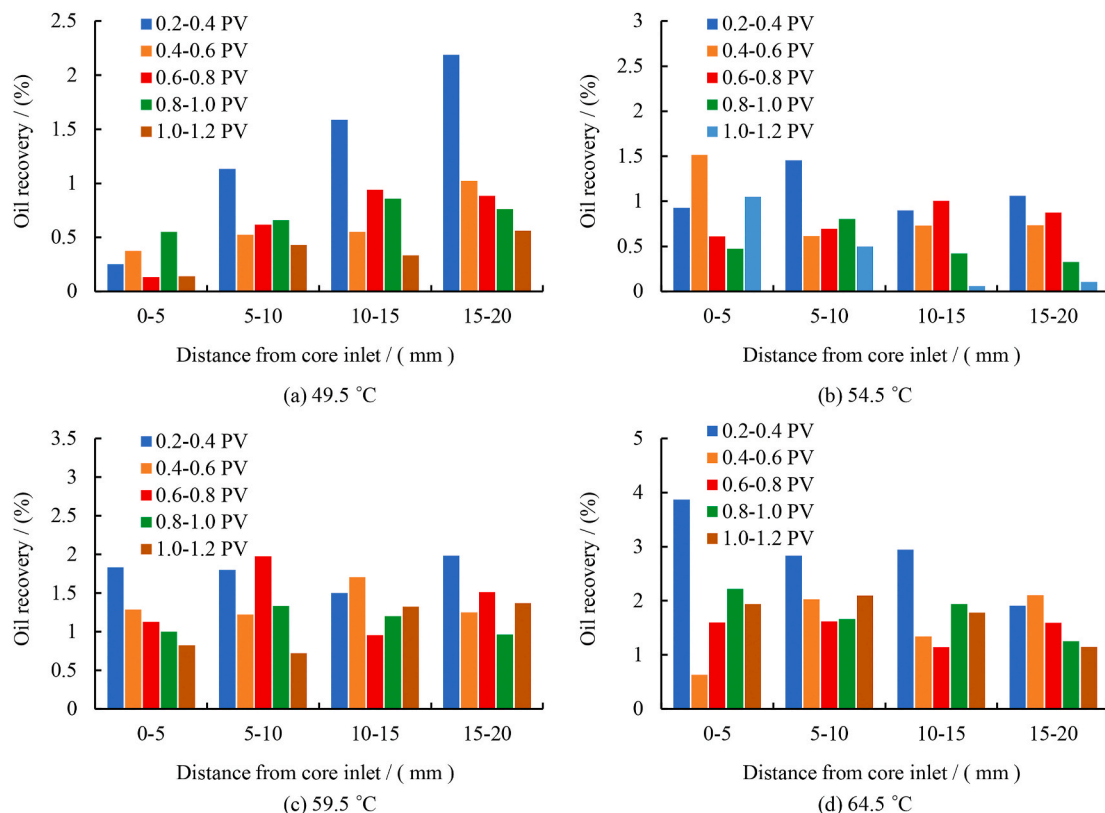


Fig. 9. Change in oil recovery under various distance ranges and PV injections at various injection temperatures.

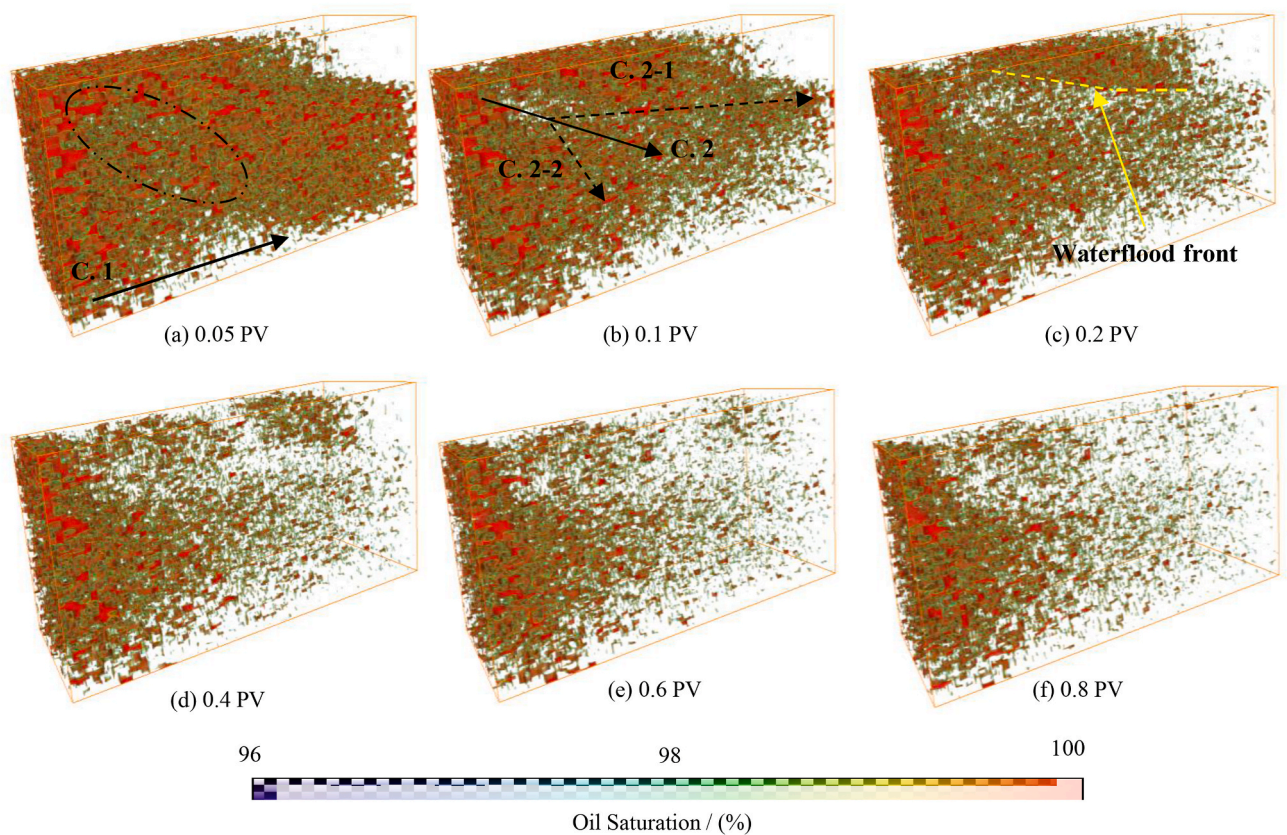


Fig. 10. Oil saturation distribution of various PV injections at water injection temperature of 54.5 °C.

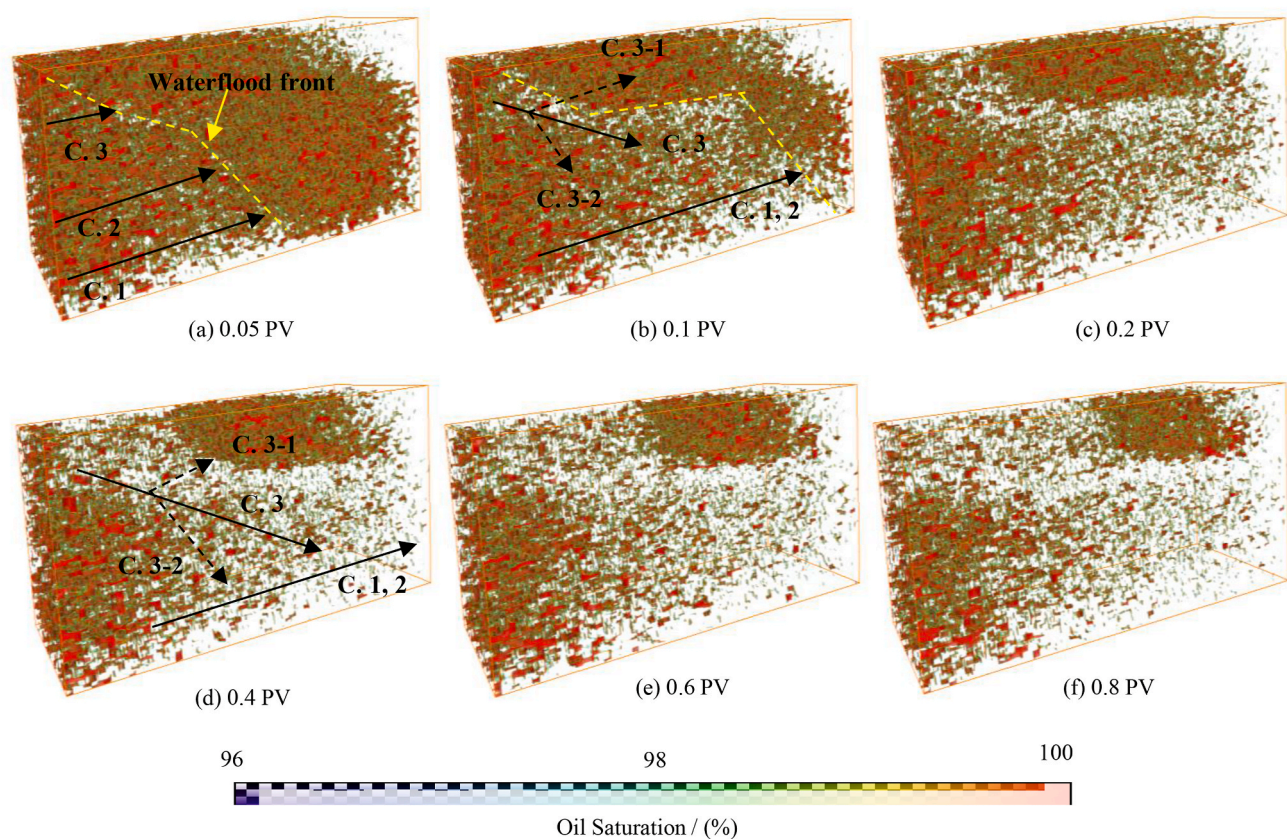


Fig. 11. Oil saturation distribution of various PV injection at water injections temperature of 59.5 °C.

visible at a 0.05 PV injection, as shown in Fig. 11a. Subsequently, with a 0.1 PV water injection, the water-flood front in the 700-mD and 1200-mD layers approaches the outlet, whereas the water-flood front in the 100-mD layer has just left the inlet. As the migration rate of heavy components in crude oil is slower than that of light components, the wax barrier formed by the heavy components and wax crystal particles near the inlet constrains the downward movement of the injected water in the upper core part. In a certain range, the flow channel #3-1 (C. 3-1) dominates.

However, when the injection volume reaches 0.4 PV, the effect of gravity is aggravated, owing to the weakening of the barrier effect on the side of the core outlet. The downward movement of the top injection water converges on the injection water in the medium-high permeability layer, and flow channel # 3-2 (C. 3-2) becomes the main flow channel in the 100-mD layer. Therefore, the oil utilization degree of the 100-mD layer near the inlet is better than that near the outlet, but the overall utilization degree remains weaker than that at 54.5 °C.

As for the water injection temperature of 64.5 °C, the formation and development of the flow channel is basically consistent with that of a conventional positive-rhythm reservoir (Fig. 12). Because there is no wax deposition at the inlet to block the pores, the injected water enters the middle and high-permeability layers first, owing to the influence of the longitudinal heterogeneity, showing a typical piston-like displacement. The oil-water interface is significant, and the water-flood front in the 1200-mD layer is faster than that in the 700-mD layer. Because there is no barrier formed by wax deposition, the injected water in the 100-mD layer flows for a short distance along the core axial direction, and then converges on the injected water in the middle and high-permeability layers under gravity effects. When the water injection volume reaches 0.8 PV, some crude oil remains at the inlet. The migration speed of the heavy components is lower than that of the light components, leading to an increase in viscosity and a decrease in displacement efficiency. At the higher displacement rate in the later stages, the injected water moves

along the main channel direction under the action of a larger inertial force, so it has difficulty entering smaller pores for displacement.

3.2.2. Oil saturation profiles in different layers

From Fig. 13 to Fig. 16, it can be seen that there are three types of oil saturation distributions: the piston-like displacement type, water-flooding breakthrough type, and mixed type of the former two. The characteristic of the piston-like water flooding is the evident oil-water interface, i.e., there is an interface perpendicular to the core axial direction and moving toward the outlet in the oil saturation distribution (Fig. 14a and Fig. 16b). Because an artificial core is used, the micro-heterogeneity is weak, and the front edge distribution of the oil saturation is also more concentrated (Leng et al., 2013; LÜ et al., 2013) (see Fig. 15).

The water-flooding breakthrough type is characterized by no evident oil-water interface in the vertical direction. This is mainly because the injected water displaces along a certain section of the layer instead of the entire layer, and forms a breakthrough channel. Hence, there is no abrupt rise or fall in a certain section in regards to oil saturation (Fig. 13a).

The mixed type has its own characteristics, which appear at low water injection temperatures. In the early stages, the water-flood breakthrough type appears, and in the later period, an approximate piston-like displacement dominates. However, because water breakthroughs in some flow channels result in a decrease in oil saturation, the distribution of the oil saturation front is gentler (Figs. 13c and 14a), and the slope is smaller than that of the piston-like displacement type.

Hence, when the water injection temperature is 49.5 °C, the 100-mD layer is dominated by a water-flooding breakthrough; the 1200-mD layer presents a mixed type, but the water breakthrough impacts the most. The change in the oil saturation profile in the 700-mD layer is more complicated. The oil saturation is higher than in the initial condition at 0.05 PV, 44 mm away from the inlet, and at 0.1 PV, 67 mm

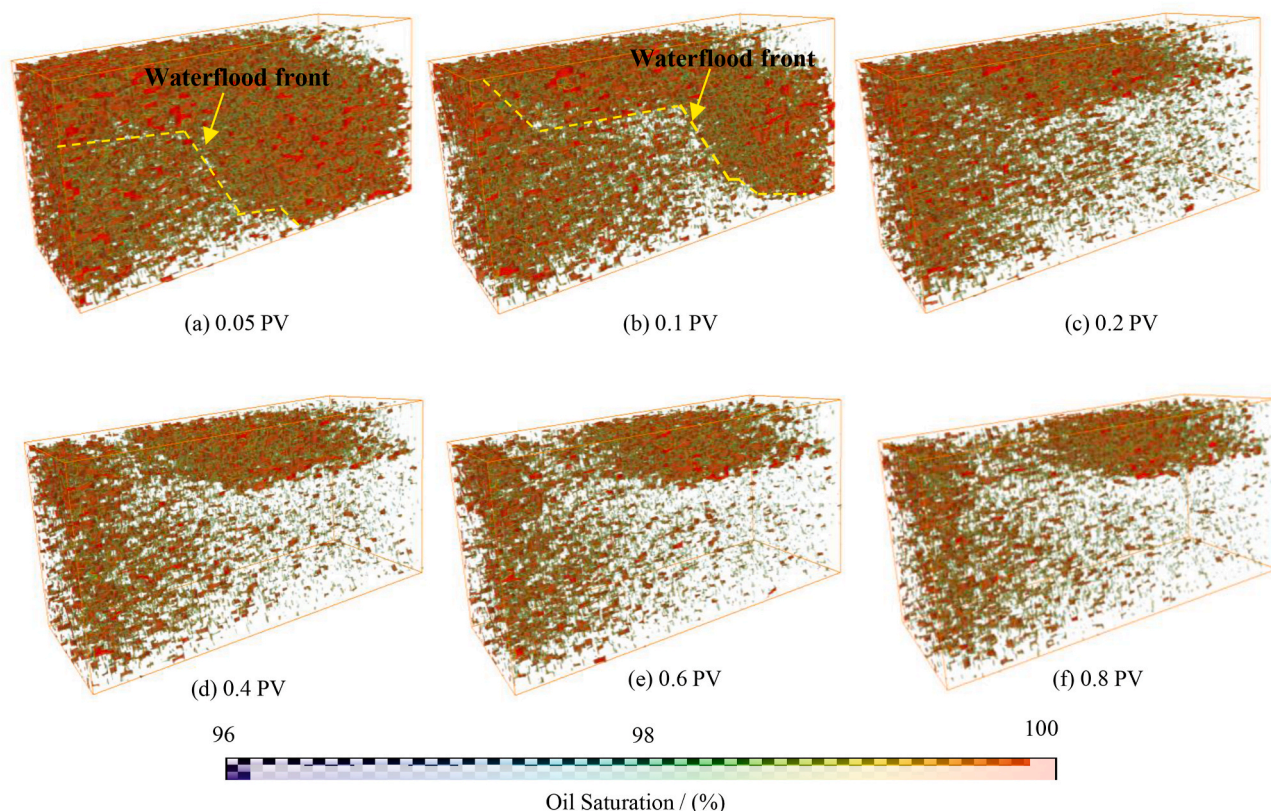


Fig. 12. Oil saturation distribution of various PV injections at water injection temperature of 64.5 °C.

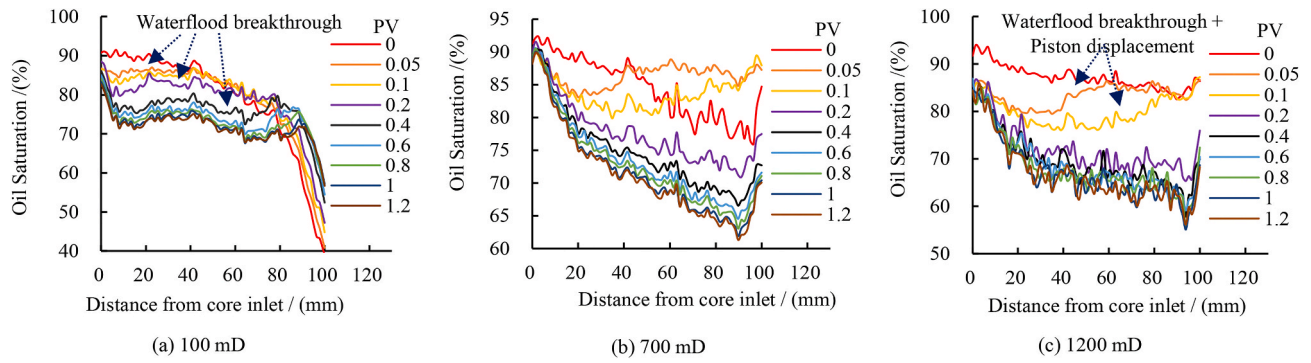


Fig. 13. Oil saturation profile under various PV injections in different layers at injection temperature of 49.5 °C.

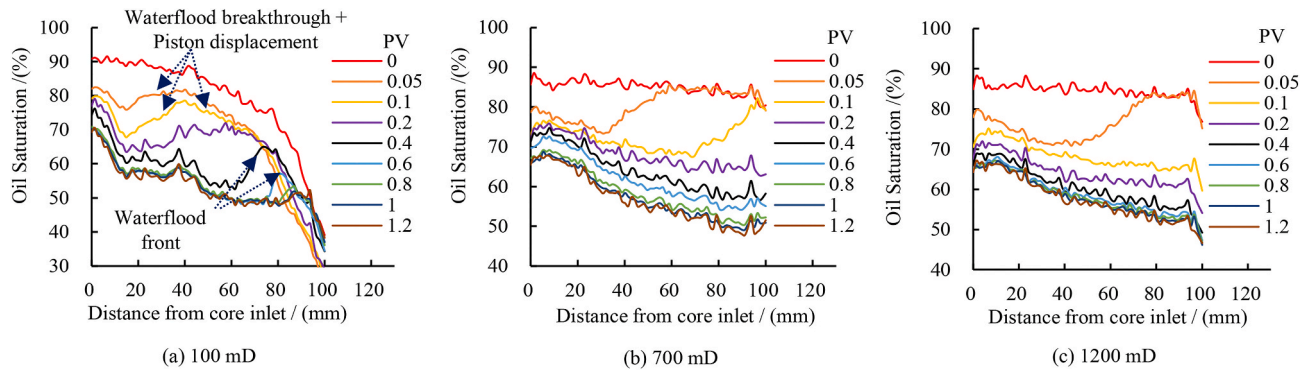


Fig. 14. Oil saturation profile under various PV injections in different layers at injection temperature of 54.5 °C.

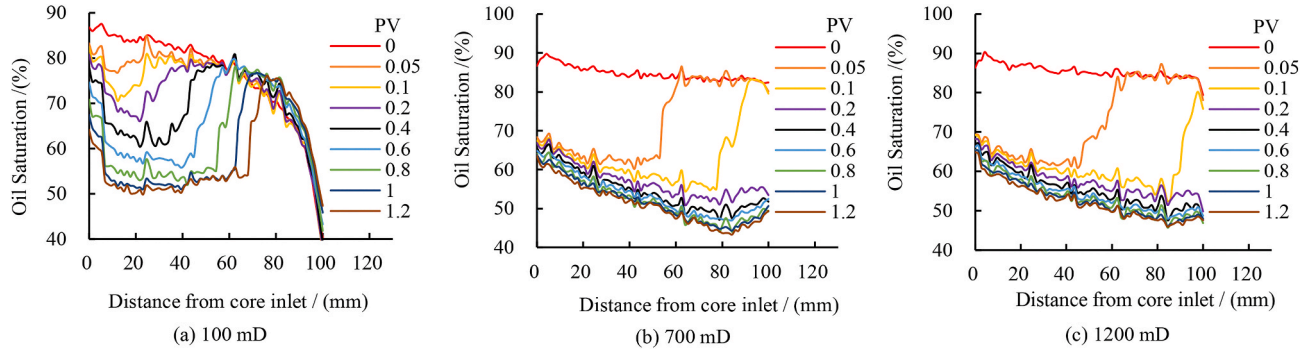


Fig. 15. Oil saturation profile under various PV injections in different layers at injection temperature of 59.5 °C.

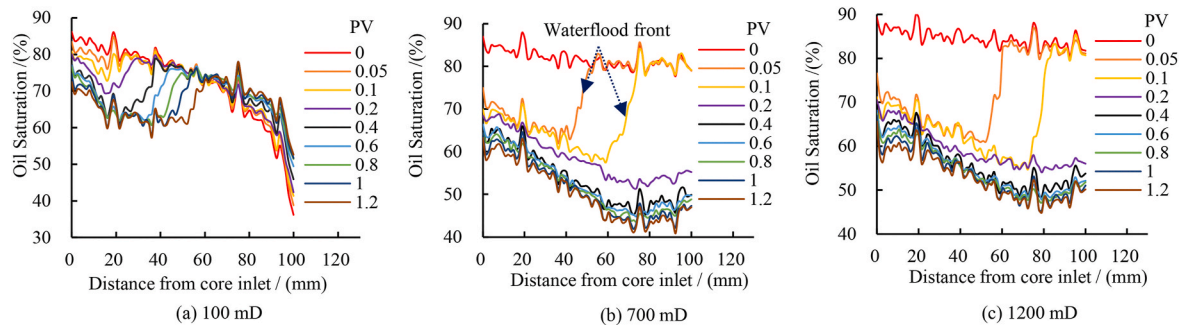


Fig. 16. Oil saturation profile under various PV injections in different layers at injection temperature of 64.5 °C.

away from the inlet. This may be because the lower-temperature injected water cools the crude oil near the oil-water front edge and squeezes the surrounding oil when displacing along the bottom of the 1200-mD layer; thus, part of the crude oil is diverted to the 700-mD layer. Only when the injected water breakthrough forms a flow channel is the core oil saturation significantly reduced, and there are differences in the timing of different layer declines. The higher the permeability, the earlier the reservoir declines. Therefore, we conclude that for ultra-low temperature water injection, the injected water breakthrough is conducive to expanding the sweeping area and improving the recovery rate. The higher the permeability of the reservoir, the earlier the breakthrough and flow channel of injected water in the next permeability reservoir.

At the 54.5 °C injected water temperature, the oil saturation distribution curve of the middle and high-permeability layers shows a clear mixed type, and the piston-like displacement type at end of the curve is significantly better than that at 49.5 °C. As shown in Fig. 14a, in the early stage, the curve type in the top 100-mD layer is dominated by a mixed type, but subsequently, it has evident piston-like displacement characteristics. Therefore, the displacement effect of the injected water in the 100-mD layer at 54.5 °C is significantly better than that at other temperatures.

When the water injection temperature is further increased to 59.5 °C or 64.5 °C, the displacement characteristics of the piston-like displacements in the middle and high-permeability layers are significant. The oil saturation distribution curve of the 100-mD layer shows a composite shape in the early stage, and soon turns into a piston-like displacement. In the 100-mD layer at the start of displacement, the injected water rate is lower, and the injected water always breaks the path of least resistance. However, when the injected water in multiple pores converges to form a flow channel, the seepage resistance decreases, the partial flow increases, and the piston-like displacement begins to appear. In addition, because there is no wax deposition barrier, the oil saturation in the 100-mD layer no longer decreases significantly after 60 mm away from the inlet at 64.5 °C, which is much smaller than the 89 mm at 54.5 °C, and the 74 mm at 59.5 °C. Therefore, within a certain range of water injection temperatures, moderate wax precipitation causes the cold water-flooding to show profile-control characteristics; the sweeping area is expanded, and oil recovery is improved in the top low-permeability layer. However, in this temperature range, the microscopic oil displacement efficiency becomes worse than at high temperatures, and the oil recovery decreases.

3.2.3. Change in pressure drop

The analysis of pressure drop at both ends of the 10 cm core is another method for evaluating the wax deposition area at the inlet and development of the flow channel. Fig. 17 shows that the pressure drop

curve of 49.5 °C can be divided into three stages. The first stage indicates wax deposition and water breakthrough at the inlet. For high-PPT oil, the amount of wax deposition is affected by the temperature and pressure (Yong and Llave, 1996; H. Chen et al., 2015). A large temperature difference between the water injection temperature and reservoir temperature will lead to a rapid decrease in the crude oil temperature; in addition, the growth rate of the wax crystals is faster than the nucleation rate (Ekweribe et al., 2009) and the sizes of the precipitated wax crystal particles are larger, making it easier for them to be trapped and blocked by smaller pores. Moreover, there is a positive correlation between pressure and wax precipitation. The increasing pressure results in an increase in the WAT, and an accelerating wax precipitation rate. This will constitute a cycle of chronic deterioration. The higher the injection volume, the higher the pressure, and the more evident the blocking degree. Therefore, in the first stage, the pressure drop increases sharply with increasing water injection. The maximum pressure gradient can reach 22.73 MPa/m at a 0.1 PV injection. Subsequently, the injection water breakthrough at the bottom of the 1200-mD layer causes the pressure to decrease, and the closely following small wave peak reflects the formation of the flow channel 2 in the upper core.

In the second stage, the displacement pressure drop fluctuates between 1.5 MPa and 1.8 MPa, indicating that the injection water is expanding the sweeping area near the flow channel. The injected water flushes, denudes, and displaces the crude oil in the pores or the wax crystal blocked in the pores at a high flow rate. When the injection volume reaches 0.4 PV, in the third stage, after a short and rapid decline, the pressure drop enters a gentle decline stage. The injected water mainly moves along the flow channel, and the sweeping area is no longer further expanded.

However, with increasing water injection temperature, the duration of the wax deposition stage is shortened, or even disappears. The pressure drop curve at 54.5 °C comprises only two stages (Fig. 18). The first stage is similar to that at 49.5 °C, and indicates the blockage of the inlet and the injected water breakthrough. However, after the end of the first stage, the displacement pressure drops rapidly and then tends to be gentle; there is no pressure fluctuation period. This is owing to the decreasing plugging degree of the pore and throat caused by increasing the injection water temperature. Fig. 19 shows that the wax deposition plug effect of the first stage is shorter at 59.5 °C; the displacement pressure drop increases rapidly and then decreases rapidly, and the pressure drop curve shows a downward fluctuation trend. At this time, the large wax crystal particles formed at the inlet move continuously under the motion of the fluid, are deposited at the pores to form blockages, and then break and continue to move under a larger pressure drop.

At 64.5 °C, the maximum pressure difference is only 0.128 kPa (Fig. 20). The pressure drop fluctuation displays the Jamin effect when

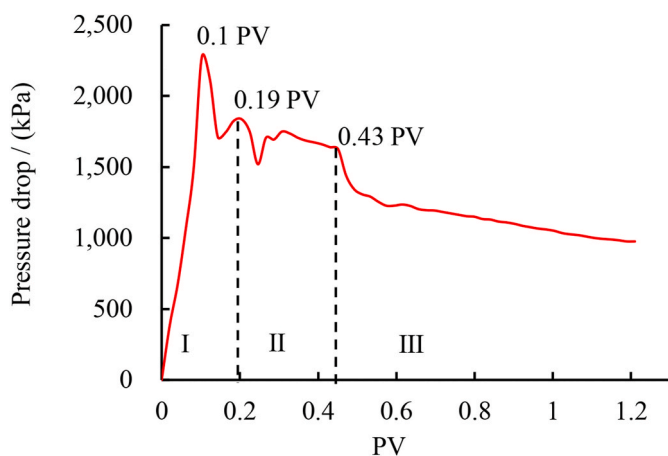


Fig. 17. Pressure drop curve at water injection temperature of 49.5 °C.

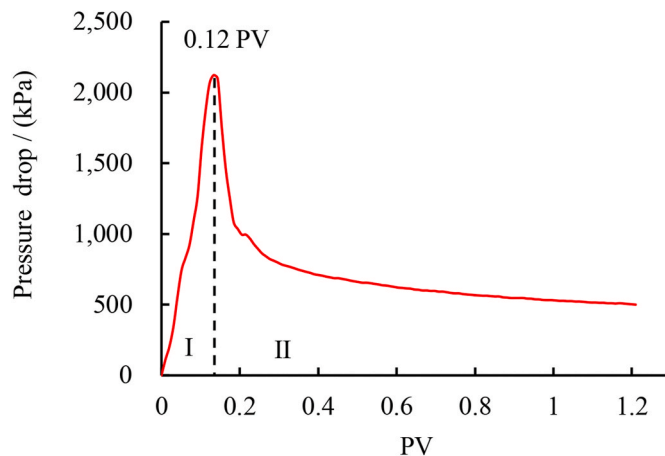


Fig. 18. Pressure drop curve at water injection temperature of 54.5 °C.

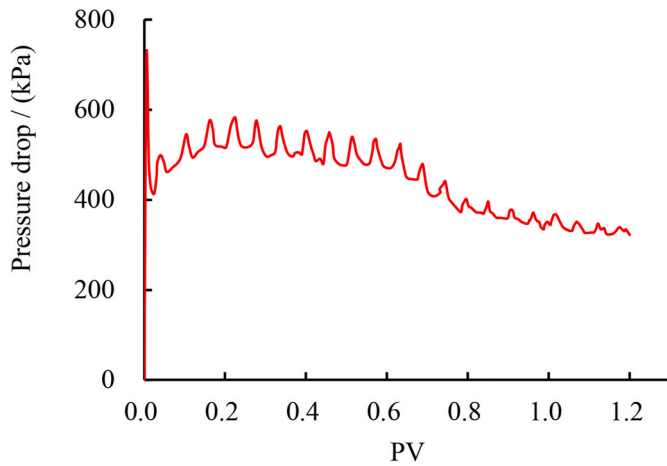


Fig. 19. Pressure drop curve at water injection temperature of 59.5 °C.

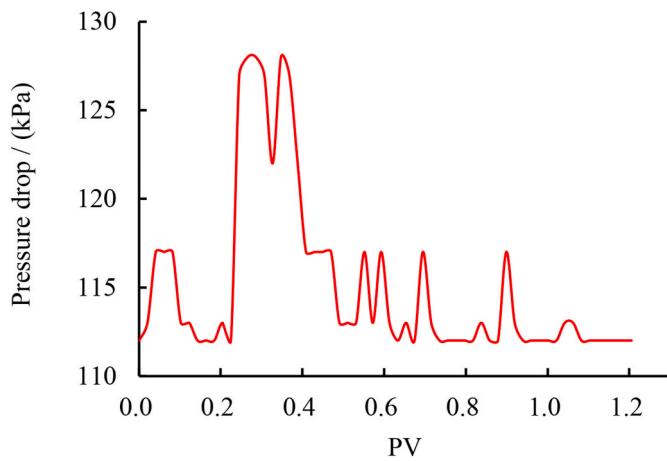


Fig. 20. Pressure drop curve at water injection temperature of 64.5 °C.

the injected water breaks through the throat or from the migration of solid particles when the clay particles expand. Finally, when the displacement pressure drop tends to be stable, the displacement pressure drop increases with decreasing temperature, i.e., to 112 kPa, 322 kPa, 501 kPa, and 976 kPa, respectively.

3.2.4. Displacement model for high-pour-point oil

Based on the above results, the displacements from injected water at different temperatures in high-PPT oil reservoirs with positive rhythm characteristics can be divided into three types: water-flooding breakthroughs, approximate-equilibrium piston-like displacements, and non-equilibrium piston-like displacements. Different displacement types

have different characteristics.

The main feature of the water-flooding breakthrough type is that the water injection breakthrough is followed by oil displacement (Fig. 21 a). At low temperatures, layers with different permeabilities have strong resistance to the flow of the injected water, forming an elliptical wax deposition area at the entrance and wax deposition barrier at the far end; among these, the middle-permeability layer is the most serious. The low temperature causes strong wax deposition in the oil-water front to inhibit the flow of injected water, forcing cold water to flow to areas with less resistance, and then forming a new wax deposition area. Owing to the lower resistance of the high-permeability layer, the larger pore size reduces the ability to capture the wax crystal particles. Simultaneously, the higher flow rate increases the ability to destroy the wax network structure, so that the paraffin molecules in the high-permeability layer do not have enough time for crystallization. Therefore, the cold water gradually expands the range after the high-permeability layer is broken through. In addition, the movement speed of the cold water in the high-permeability layer is faster than that in the medium-permeability layer. This expands the sweep range, and reduces the temperature of the areas not swept by the injected water in the medium-permeability layer owing to the heat transfer; then, paraffin deposition occurs. This is the main reason for the formation of the wax deposition barrier, but it is difficult to further transform it into a low-permeability layer. The serious wax deposition increases the flow resistance of the middle-permeability layer and may exceed that of the low-permeability layer, forcing the injected water to flow in the low-permeability layer with less resistance. Meanwhile, the cold water in the top low-permeability layer moves towards the middle-permeability layer under the action of gravity, and continues to promote the formation of the wax deposition barrier. When the injection water breaks through to form flow channels, the injected water erodes the wax and crude oil near the flow channels at a higher flow rate, expanding the sweeping area and increasing the oil recovery.

The main characteristics of the approximate-equilibrium piston-like displacement type are that the injected water in each layer flows approximately evenly, and that the sweeping area of the low-permeability layer is significantly improved (Fig. 21b). This situation mainly occurs when the water injection temperature is slightly lower than that of the WAT. The wax deposition area formed by the injection water at the inlet is reduced; the thermal convection causes the injection water temperature to rise away from the wax deposition area at the inlet, and the medium-to-high permeability layer exhibits piston-like displacement characteristics. Meanwhile, the wax deposition barrier constrains the downward movement of the top-injected water under the influence of gravity. The injected water in the low-permeability layer also exhibits piston-like displacements with increasing PV injection, and the flow rate is just slower than that in the middle-high-permeability layer. Hence, an approximate-equilibrium displacement is formed in the reservoir, and the oil recovery of the low-permeability layer is significantly improved.

The main characteristic of the non-equilibrium piston-like

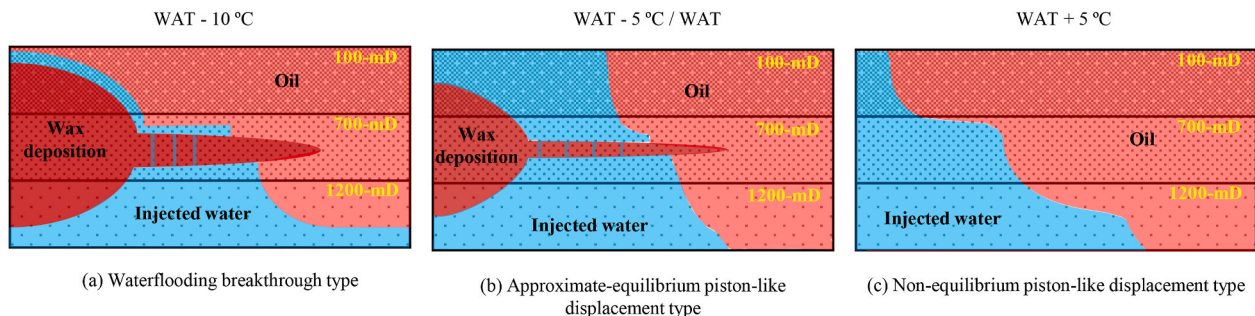


Fig. 21. Schematic diagram of displacement mode.

displacement type is that the higher the layer permeability, the faster the injected water flow rate, and the lower the oil recovery in the top low-permeability layer (Fig. 21c). There is no wax deposition in the pores, the water injection process is similar to that in conventional reservoirs, and the heterogeneity in the layer leads to a significant decrease in the utilization of the top low-permeability layer. Notably, Fig. 21 is only a schematic diagram of the injected water sweep range and does not represent the true shape of the injected water front. Finally, there will be large differences in the microscopic oil displacement efficiency within the sweeping area at different injection temperatures.

3.3. Displacement efficiency and trapped oil distribution

3.3.1. Heavy components retention in pores

It is important to study the retention of heavy components in the reservoir. According to the development history of the S84A12 high-PPT oil reservoir, the WAT of the dead oil became higher than that in the initial stage, by more than 6.6 °C. Wax deposition and the slow migration of heavy components were the reasons for the retention. Hence, in this study, 0.6 PV and 1.2 PV injections are used as the cut-off points; the oil samples are collected from the outlet, the saturation and wax contents are tested, and the results are compared with the values at the initial time for analysis. According to the standard, the wax is obtained by removing a polar substance by alumina column chromatography, and then crystallizing it at -20 °C with a benzene-acetone (1:1 by volume) solution as a dewaxing agent. The saturated content is obtained by precipitating and filtering through n-heptane precipitation, followed by evaporation of the n-heptane solution in a vacuum box. The large content of wax and saturation in high-PPT oil made us choose them as the analysis objects.

First, wax is the main component of saturation, and is greatly affected by temperature. The saturated and wax contents account for a large proportion of the original crude oil, reaching 84.17% and 55.18%, respectively, and the changes in their mass fractions generally represent their true change trends. We cannot calculate the actual retention of the related components in the core, i.e., the saturated and wax contents in the original oil sample and filter oil sample cannot be directly reduced. The most significant problem is that the crude oil quality scale is not the same during the tests.

Fig. 22 and Fig. 23 show that the saturated and wax contents in the collected oil sample are less than those of the original oil sample, regardless of the injection temperature, indicating that both are partially retained in the rock pores. When the water injection temperature is higher than WAT, the wax mass fraction in the crude oil at a 1.2 PV is higher than that at a 0.6 PV, i.e., the content of heavy components in the oil sample is increased. This is the most direct evidence of the slow

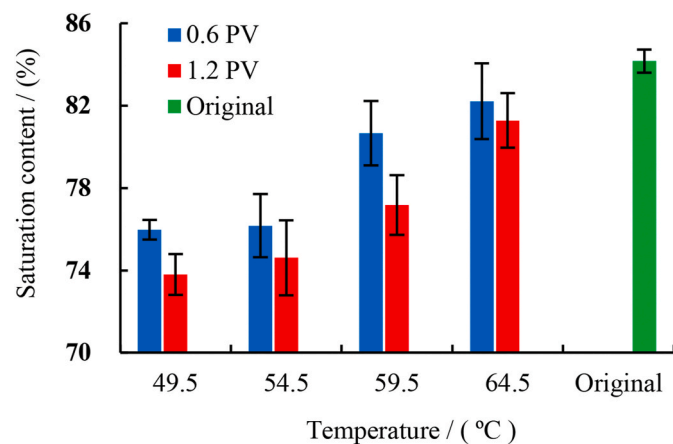


Fig. 22. Change of saturated content in oil samples collected under various PVs of water and temperature.

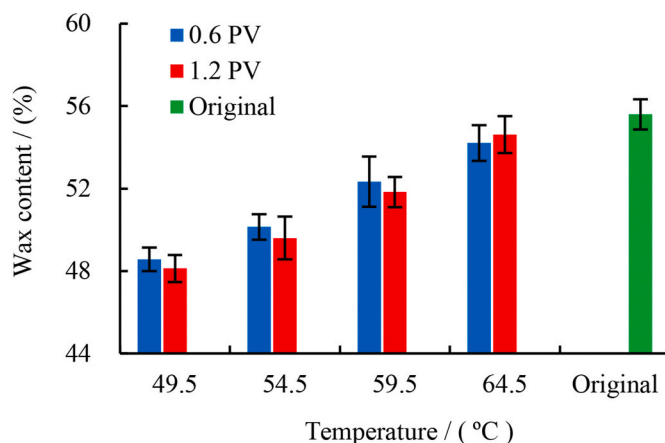


Fig. 23. Change of wax content in oil samples collected under various PVs of water and temperature.

migration rate of the heavy components. The increase in the heavy component content in the reservoir will lead to increases in the oil viscosity and WAT, and a decrease in the displacement efficiency. These characteristics are consistent with the oilfield production results.

When the wax crystals are precipitated from the crude oil, the saturation and wax contents show downward trends with increasing PV injections. This shows that the wax deposition has become the main reason for the crude oil retention. In particular, wax crystals with a high carbon number are precipitated, and form a spatial network structure to bind the crude oil in it and constrain its further flow. The precipitated wax crystal particles aggregate at the hole to form a blockage. Part of the wax will be deposited on the pore surface, changing the wettability of the reservoir from water to oil, and the effect of the capillary force will be weakened. Therefore, for the development of high-PPT oil reservoirs, it is necessary to regularly analyze and compare the wellhead crude oil components and physical properties, and to adjust the water injection temperature and development schemes to reduce the influence of heavy component content retention.

3.3.2. Displacement efficiency

Fig. 24 shows the oil recovery rate of each layer. The oil recovery of the middle- and high-permeability layers is better than that of the top 100-mD layer; the oil recoveries of the middle- and high-permeability layer mostly show declining trends with decreasing water injection temperatures. In the approximate-equilibrium piston-like displacement model, the oil recoveries of 100-mD layers at 54.5 °C and 59.5 °C are significantly better than those at other temperatures, reaching 18.2% and 16.7% higher, respectively, than the 4.9% at 49.5 °C and 9.8% at

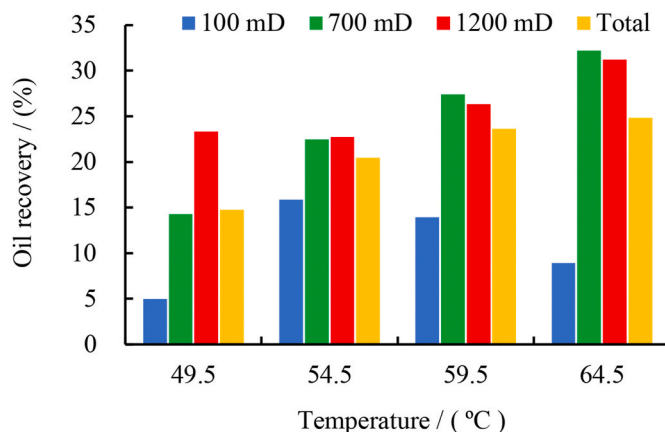


Fig. 24. Oil recovery in different layers (CT).

64.5 °C. Therefore, we can conclude that moderate wax precipitation is beneficial to improving the oil recovery of the top low-permeability layers.

In addition, at 59.5 °C and 64.5 °C, the oil recovery of the 700-mD layer is slightly higher than that of the 1200-mD layer. The reason for this may be as follows. Owing to the disappearance of the wax deposition barrier, fluid convection occurs between the 700-mD and 1200-mD layers. The water injected into the 100-mD and 700-mD layers at the top drives part of the crude oil down into the 1200-mD layer under the action of gravity (although the exchange volume cannot be measured yet). Moreover, the iodoheptane retention may lead to an increase in the CT value. As there is no wax precipitation, the high-density iodoheptane stays in the pores with the heavy components, resulting in an unreasonable increase in the CT values in the pixel units. As shown in Fig. 25, the difference between the real core oil recovery and oil recovery calculated using the CT values increases with increasing water injection temperature. Further, the migration of the dispersed crude oil may also lead to an increase in the CT value. In the high water-cut stage, the crude oil (as the dispersed phase) in the 100-mD and 700-mD layers is carried by the injected water from the pores, or is retained in the small pores. At this time, although the water phase in the pores is the continuous phase and occupies most of the pores, a small amount of crude oil added to the CT enhancer can cause an unreasonable increase in CT value. The layer with the strongest water washing degree is most affected by the CT value of the dispersed-phase crude oil; therefore, the calculated crude oil recovery is relatively small. Unfortunately, although we considered the influences of the temperature change on the crude oil CT values before the experiment and corrected the CT_{oil} values at different temperatures, there are still other factors affecting the results. However, the fluctuation of the CT value does not affect the analysis regarding the formation and development of the flow channel.

Finally, the recovery rate increases with increasing water injection temperature, but the rate of improvement slows down (Fig. 25). In the early 0.2 PV injection, more than 50% of the crude oil is produced. The effect of temperature is consistent with the expected results. The recovery reaches 36.8% at 64.5 °C, and 17.6% at 49.5 °C. However, in different displacement and temperature ranges, the changes in the oil recovery rates are different. When the water injection temperature is above 54.5 °C, the oil recovery growth rate in the first 0.2 PV is significantly higher than that at 49.5 °C; however, when the water injection volume continues to increase, a recovery difference gradually appears between the temperatures. Therefore, it is the most effective measure to maintain the injection temperature above the WAT in the early stage.

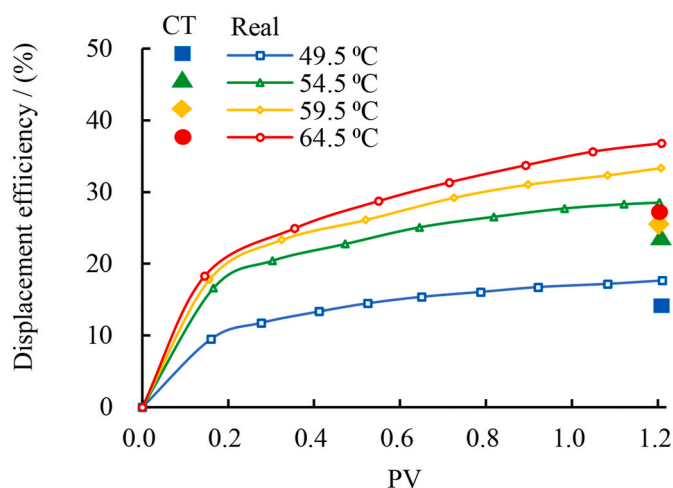


Fig. 25. Displacement efficiency of different water injection volumes under different temperatures (experiment).

3.3.3. Distribution characteristics of trapped oil

At a 1.2 PV injection, the water-cut reaches more than 99%, and different types of trapped oil are formed at different positions. Among them, some trapped oils are formed at a specific water injection temperature. Hence, based on the difference between the three displacement modes, oil saturation slices along the core axial direction at 49.5 °C, 54.5 °C, and 64.5 °C are selected for analysis. As shown in Fig. 26, the trapped oil distribution can be divided into three types, namely, trapped oil formed by wax deposition, trapped oil formed by weak water washing, and trapped oil formed by reservoir rhythm changes in the layer.

The trapped oil formed by wax deposition is mainly caused by low temperatures. The precipitated wax crystals tend to gather and form a network structure, trapping liquid oil inside and forming a gel state. The injected water in the wax deposition area causes the cooling effect, invades the crude oil's original space, and promotes the gelation process (Sun et al., 2016a,b). That is, the critical wax precipitation amount causing the gelation of the crude oil is reduced. This gelation can only be destroyed by a significant pressure difference. This phenomenon is mainly formed at the inlet with an ultra-low temperature water injection, such as the dark red area concentrated at the inlet in Fig. 26a. When the injection temperature increases, the wax precipitation decreases, leading to a decrease in the strength and network structure size. It shows a disperse phase under long-term injected water flushing, so the wax deposition area at the inlet decreases at 54.5 °C. In addition, there are also trapped oil concentration areas formed by wax depositions near the main flow channel, such as that in the 100-mD layer in the upper part of the reservoir, and the wax deposition barrier in the core. However, these two trapped oils mainly appear at extremely low-temperature water injections, and gradually disappear or change into trapped oils formed by other factors with increasing water injection temperatures (Fig. 25b).

The trapped oil formed by a weak water washing degree is mainly caused by changes in fluid physical properties. An increase in the heavy component content in the reservoir makes the fingering phenomenon more evident. Simultaneously, some of the heavy components are adsorbed on the pore and throat surfaces, changing the wetting index, and reducing the role of the capillary force in microscopic displacement. Then, when the cold water flows away from the inlet through the pores containing oil, a wax film will also be formed on the oil-water contact surface, increasing the flow resistance. Further, once the main flow channel is formed, the injected water (with a lower viscosity) has a greater inertial force at a high flow rate, and moves toward the outlet. It is difficult to obtain the lateral angular velocity into the pore to displace the crude oil. In addition, because an artificial core is used, the effect of the micro-heterogeneity on the distribution of the trapped oil can be ignored in the experiment, but the effect of micro-heterogeneity in real cores is important. Moreover, there are island-like distributions of trapped oil in the core away from the inlet; and as shown by the dashed circles in Fig. 26b and c, the higher the injection temperature, the more evident that the injected water flows past the islands.

The trapped oil formed by reservoir rhythmic changes is mainly caused by the heterogeneity in the layer and the difference in oil-water density. However, this mainly occurs when the water injection temperature is higher than the WAT. Owing to the positive rhythmic distribution of the core permeability and difference in the viscosity and density values of oil and water, the front edge of the water drive oil rushes to the bottom of the layer at a certain distance from the injection inlet, resulting in a considerable part of the oil in the upper 100-mD layer not being swept (Fig. 26c).

3.4. Further discussion on wax deposition

In summary, both the reservoir heterogeneity and water injection temperature have a significant impact on oil recovery. Therefore, for the subsequent development of the S84A12 high-PPT reservoir, the most important thing is to reduce the adverse effects of wax deposition based

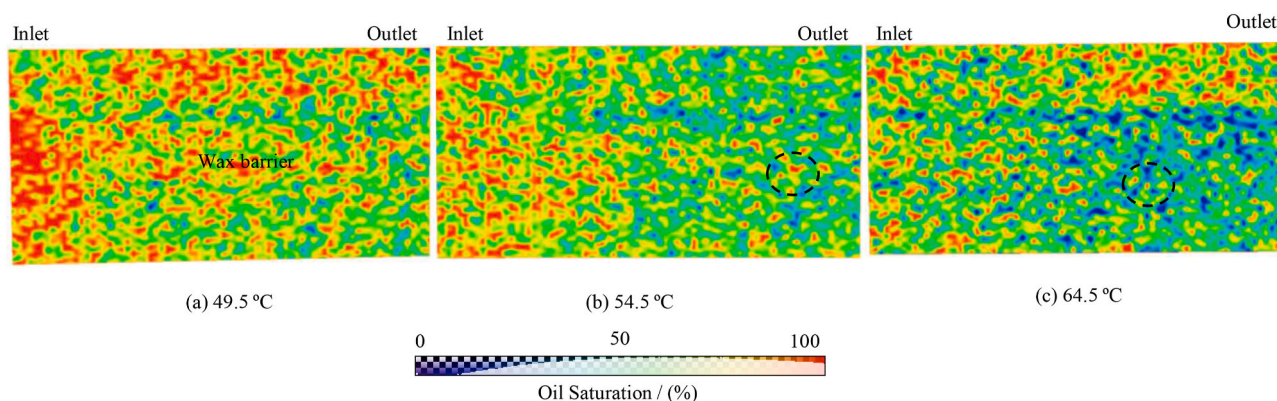


Fig. 26. Oil saturation slice at 1.2 PV injection at various temperatures (The dashed circles are the island-like remaining oil).

on improving the fluidity profile, while simultaneously improving the microscopic oil displacement efficiency. From current laboratory experiments and application experience, it is shown that using only hot water or hot steam injection after cold water flooding has a higher cost and lower efficiency. ASP compound flooding can improve the microscopic oil displacement efficiency, while blocking the main flow channel. In addition, for injection wells that already have formation damage and a sharp increase in the injection pressure, solving the near-well blockage should be the first step. The unfavorable effects of serious wax deposition in the reservoir can be solved by increasing the water injection temperature, adding a certain amount of paraffin inhibitor, or injecting a self-heating chemical slug.

There are still many limitations to our work. First, NaI was added to the water instead of NaBr, which may have reduced the impact of the NaBr deposition on the evaluation of oil recovery in the high-permeability layer. The degree of deposition of NaI in the reservoir and its influence on the physical properties of the ASP flooding solution should be further explored. Second, the WAT and wax precipitation rate of the crude oil should be considered in the process of decreasing pressure (M. Wang et al., 2020). At the same water injection temperature, a reduction of the formation pressure helps to reduce the adverse effects of wax deposition. To achieve the same amount of wax deposition, a lower-temperature injection water and larger injection volume may be required under low-pressure conditions. In addition, the core of this study is an idealized simulation of the heterogeneity of the actual reservoir. In fact, the heterogeneity of the reservoir, sedimentary rhythm, and distribution of real core pores will have varying degrees of influence on wax deposition and flow channel formation. Wax deposition and its influence in porous media is an extremely complex process. The temperature, pressure, crude oil composition, injection medium, etc. will all affect it. The above questions will be answered in future works. The focus of our next study is a quantitative characterization of the degree of wax deposition in a porous media wall, and the resistance formation mechanisms during flows of high-PPT oil in porous media.

4. Conclusion

The wax deposition caused by cold water injection has been a long-term challenge to enhancing oil recovery in high-PPT oil reservoirs. We conducted an on-line CT study to investigate the formation and development of a flow channel, heavy component retention, displacement efficiency, and trapped oil distribution characteristics in a special reservoir under various water injection temperatures. This is the first study to investigate the influence of the wax deposition on the flow channel. The research results also provide initial insights for evaluating the cold damage of high-PPT oil, and provide theoretical support for a subsequent ASP recovery enhancement technical scheme. The conclusions are as follows.

There is an evident heavy component retention phenomenon in the development of a high-PPT oil reservoir, and it increases gradually with decreasing injection water temperatures. The middle-permeability layer has the most serious wax deposition at the inlet. The displacement models for injected water with different temperatures in a reservoir with a positive rhythm can be divided into three types: a water-flooding breakthrough model, approximate-equilibrium piston-like displacement model, and non-equilibrium piston-like displacement model. Different displacement types have different characteristics.

The water-flooding breakthrough model is mainly formed when the water injection temperature is far lower than the WAT, and has the characteristics of an initial water breakthrough followed by oil displacement. There are double channels in the flow profile, and the wax deposition barrier is between them. The formation time of the upper flow channel is evidently later than that of the bottom high-permeability layer. The pressure drop and PV curve show a three-stage distribution; the oil saturation distribution curve types are the water-flooding breakthrough type and mixed type. When the water injection volume is greater than 0.4 PV, there is an improvement in the oil recovery caused by the vortex phenomenon at the inlet in the bottom high-permeability layer. The trapped oil is mainly distributed at the inlet and the upper part of the reservoir, and is mostly caused by wax deposition.

The approximate-equilibrium piston-like displacement model mainly occurs when the water injection temperature is between the WAT and 5 °C below the WAT. The significant characteristic is that the moderate wax precipitation increases the oil recovery from the upper low-permeability layer. The dual channel and wax deposition barrier of the flow profile gradually disappear with increasing injection temperatures. The heterogeneity of the wax barrier distribution results in a better oil recovery at 54.5 °C at the top of the low-permeability layer than that at 59.5 °C; the oil recovery at the inlet is better than that at the outlet at 59.5 °C. The pressure drops and PV curve show a clear two-stage type; the oil saturation distribution curve types are the piston-like displacement type and mixed type. The trapped oils are mainly caused by wax deposition and a weak water washing degree.

The non-equilibrium piston-like displacement model is mainly formed at temperatures higher than the WAT, and its characteristics are consistent with the water flooding in a conventional reservoir. The displacement pressure drop fluctuates near 0.1 MPa; the distribution curve of oil saturation is a piston-like displacement type, and the oil-water front movement is evident. The trapped oil is mainly distributed in the upper layer, where the injected water cannot reach.

Credit author statement

Mibang Wang: Design of the experimental procedure; Formal analysis; Writing – original draft preparation. Shenglai Yang: Formulation of

overarching research goals and aims. Jiajun Li: Conduct the CT displacement experiment. Zuhao Zheng: Conduct the CT displacement experiment. Jing Wen: Investigation and financial support. Quanzheng Ma: Investigation. Qian Wang: Investigation; review & edit the preparation. Hao Chen: Language editing & review.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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